

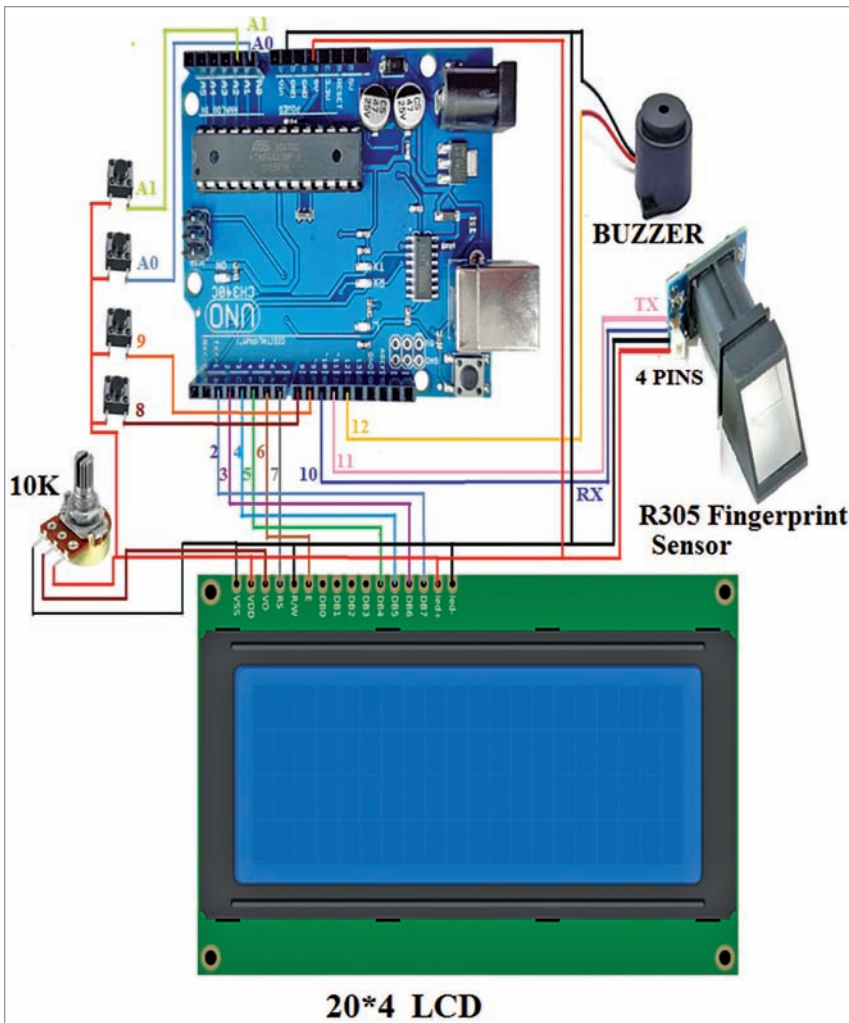


Fig. 6: Wiring diagram

voter is permitted to vote once, and the administrator does not participate in voting.

After authentication, the LCD displays:

“WELCOME (NAME) VOTE NOW...”

The voter selects a candidate using one of four push buttons connected to pins 8, 9, A0, and A1. The buzzer connected to pin 12 sounds briefly to confirm recording. The system tracks fingerprint IDs to ensure that no user votes more than once.

When all votes have been cast, the administrator authenticates to view the final results.

Working steps

1. Start. The system powers on and begins initialisation.

2. Initialisation. The LCD, fingerprint sensor (R305), buzzer, and four push buttons are initialised.

3. Fingerprint detection. The system waits for any person's (voter, candidate, or admin) fingerprint to be placed on the sensor.

4. Fingerprint match check. The system verifies whether the fingerprint exists in the stored database.

5. Admin verification. If the fingerprint matches the administrator ID (Rakesh Jain), the system displays the final vote results. If not, the system proceeds to voter verification.

6. Voter eligibility check. If the fingerprint belongs to a voter, the system checks whether a vote has already been cast by that voter.

7. Personalised welcome. If

eligible, a personalised welcome message is displayed.

“WELCOME (Voter Name) VOTE NOW...”

8. Display candidates. The LCD lists all four candidate options..

Anil, Gunmala, Rekha, and Monu

9. Vote casting. The voter selects a candidate by pressing the corresponding push button. Each candidate is assigned a dedicated button. Button 1 (Pin 8) for Anil, Button 2 (Pin 9) for Gunmala, Button 3 (Pin A0) for Rekha, and Button 4 (Pin A1) for Monu.

10. Vote recorded. The selected vote is recorded in the system. The buzzer (Pin 12) sounds for one second to confirm the vote. The voter is marked as having ‘voted’ to prevent multiple voting.

11. Ready for next voter. The system returns to fingerprint detection and awaits the next user.

12. Result display (admin only). Only the administrator is authorised to reveal the vote count by verifying their fingerprint.

13. System reset. Pressing the RESET button reinitialises the system and clears all vote records for a new session.

Software

The system operates in three main stages, with separate code for each: fingerprint enrolment, fingerprint deletion (if required), and the main voting code. Before writing these codes, the required libraries must be installed. Open the Library Manager in the Arduino IDE, search for, and install the libraries Adafruit_Fingerprint.h and LiquidCrystal.h. Once installed, coding can begin.

1. Fingerprint enrollment.

This code enrolls fingerprints into the system. When a fingerprint is being registered, the serial monitor prompts for an ID between 1 and 15. Based on this ID, a corresponding name is displayed; for example, ‘Enrolling for: Rekha.’ The user must place her finger twice to ensure accurate matching. Fig. 3 shows